

UNIT-3

COGNITIVE RADIO AND NETWORKS

INTRODUCTION

- Cognitive Radio (CR) is an intelligent wireless communication system that dynamically adjusts its transmission parameters based on the surrounding environment.
- It aims to address the growing demand for spectrum and enhance the efficient use of the radio frequency (RF) spectrum.
- The idea behind cognitive radio is to allow unlicensed users (secondary users) to opportunistically access unused licensed spectrum (called spectrum holes) without causing interference to licensed users (primary users).

SPECTRUM ACCESS MODELS

- In Cognitive Radio Networks (CRNs), the concept of spectrum access refers to how secondary users (unlicensed users) access the radio spectrum, which is primarily owned by primary users (licensed users). Since the spectrum is limited, cognitive radio technology allows dynamic sharing of spectrum between primary and secondary users. The access models define how and under what conditions secondary users can transmit on the spectrum, ensuring minimal interference with primary users.

SPECTRUM ACCESS MODELS

- **spectrum access models** are often classified into **underlay** and **overlay** models.
- how secondary (unlicensed) users access spectrum that is already being used by primary (licensed) users.
- The key difference between the two lies in the level of interference allowed and the mode of spectrum sharing.

Underlay Spectrum Access Model

- In the **Underlay Spectrum Access** model, secondary users (unlicensed users) are allowed to access spectrum bands that are already being used by primary users, but with the restriction that the interference they cause to the primary user must remain below a certain threshold. This threshold ensures that secondary users do not significantly degrade the performance of primary users.

Underlay Spectrum Access Model

- Secondary users transmit at low power levels to minimize interference with the primary users.
- The interference caused by the secondary users is continuously monitored and controlled to stay within the acceptable limit, known as the **interference temperature** or **power control threshold**.
- Under this model, secondary users typically do not need to vacate the spectrum when a primary user is active, as long as they do not exceed the interference threshold.

- **Advantages:**
- **Efficient Spectrum Use:** Secondary users can access the spectrum at all times, leading to higher spectrum efficiency.
- **No Need for Spectrum Vacating:** Secondary users do not have to leave the spectrum when the primary user is active, making it more flexible than other models.

Overlay Spectrum Access Model

- In the **Overlay Spectrum Access** model, secondary users are allowed to access the spectrum used by primary users, but they must do so opportunistically. This means that secondary users can only transmit when the spectrum is not being used by the primary users, ensuring no interference occurs.

Overlay Spectrum Access Model

- Secondary users monitor the spectrum for vacant or idle frequencies (spectrum holes).
- When a primary user is not using a specific frequency band, secondary users are allowed to use that band, but once the primary user starts transmitting, the secondary user must vacate the spectrum.
- This model typically requires **spectrum sensing** to detect when the spectrum is idle, and secondary users can only transmit when the primary user is inactive.

Overlay Spectrum Access Model

- Advantages:
- **No Interference:** Secondary users avoid interfering with primary users since they only use idle spectrum.
- **Flexible and Dynamic:** Secondary users can dynamically access different frequency bands based on availability.

Cognitive Radio Technologies and Challenges

- **1. Dynamic Spectrum Access (DSA)**
- **2. Dynamic Spectrum Sharing (DSS)**
- **3. Dynamic Spectrum Multi-Channel Operation**

Dynamic Spectrum Access (DSA)

- **Dynamic Spectrum Access (DSA)** refers to the ability of secondary users (unlicensed users) to access spectrum dynamically based on the availability of unused or underutilized spectrum, which is typically licensed to primary users (licensed users). DSA allows for the opportunistic use of the spectrum when the primary user is not actively using it.

Dynamic Spectrum Access (DSA)

- **Opportunistic Spectrum Use:** Secondary users can utilize spectrum when it is not being used by primary users, thus improving spectrum efficiency.
- **Real-time Spectrum Allocation:** DSA allows spectrum to be allocated dynamically, based on real-time spectrum availability, rather than fixed or static allocations.
- **Avoiding Interference:** DSA ensures that secondary users do not interfere with the primary users, either by vacating the spectrum when the primary user begins transmission or by controlling the interference caused by secondary users.

Dynamic Spectrum Sharing (DSS)

- **Dynamic Spectrum Sharing (DSS)** is a more specific and advanced form of spectrum sharing where multiple users (both primary and secondary) share the same frequency band, but the spectrum usage is dynamically allocated based on real-time demand, activity, or interference conditions. Unlike traditional spectrum allocation (where a band is assigned permanently to a user), DSS allows flexible, on-demand access to spectrum.

Dynamic Spectrum Sharing (DSS)

- **Real-Time Spectrum Allocation:** Spectrum bands are allocated in real-time to different users based on their need or activity.
- **Sharing Between Primary and Secondary Users:** In DSS, both primary (licensed) and secondary (unlicensed) users can share the same frequency bands, but their usage is managed to prevent interference.
- **Resource Optimization:** DSS optimizes the use of spectrum by dynamically allocating bands according to user needs, reducing wasted spectrum capacity.

Dynamic Spectrum Multi-Channel Operation

- **Dynamic Spectrum Multi-Channel Operation** refers to the ability of a cognitive radio network to operate over multiple frequency channels, dynamically selecting and switching between them based on current network conditions, interference levels, and spectrum availability. Instead of being confined to a single channel, a cognitive radio network can access multiple channels as they become available, optimizing spectrum usage across different frequency bands.

Dynamic Spectrum Multi-Channel Operation

- **Multi-Channel Spectrum Use:** The cognitive radio system can operate across multiple channels, switching between them dynamically depending on factors such as spectrum availability, interference, and bandwidth requirements.
- **Adaptive Channel Selection:** Cognitive radios adaptively select the best channel at any given time based on criteria like signal quality, power availability, or interference levels. This flexibility improves overall system performance.
- **Channel Hopping:** In some systems, cognitive radios may employ channel hopping, where they periodically switch between different channels, improving reliability and reducing the chances of interference.

The IEEE 802.22 Standard

- TV BAND AND PUS:
- The **TV bands** refer to specific portions of the radio frequency spectrum that are allocated to **television (TV) broadcasting**. These frequency bands were initially reserved for over-the-air television transmission, primarily for analog TV, but with the transition to **digital television (DTV)**, a significant portion of this spectrum became available for other uses.

TV BAND AND PUS

- **TV Band Frequency Ranges:**
- **VHF (Very High Frequency):** 54 MHz to 216 MHz
- **UHF (Ultra High Frequency):** 470 MHz to 698 MHz (this range can vary slightly in different regions)
- **TV White Spaces (TVWS):**
- **TVWS** are the gaps in the TV band that are not currently being used by licensed TV stations but are still reserved for their potential future use.
- TVWS can be utilized by **secondary users** (such as unlicensed cognitive radios) to provide broadband services, especially in rural and underserved areas where there is a lack of infrastructure like DSL or fiber optics.

Primary Users (PUs)

- **Primary Users (PUs)** refer to the **licensed users** of a particular frequency band who have the legal right to use the spectrum.
- **Licensed Spectrum Users:** Primary users are the entities with the legal right to use a specific band of spectrum. For TVWS, this includes the **TV broadcasters** using specific channels for broadcasting.
- **Protection from Interference:** The primary users must be protected from interference by secondary users (i.e., unlicensed users). Secondary users are not allowed to interfere with the primary user's transmission.
- **Priority Access:** PUs have priority over spectrum usage, meaning that secondary users (secondary cognitive radio users) must vacate the spectrum or cease transmissions if primary users begin using the band.
- **Fixed or Dynamic Use:** PUs may use their allocated spectrum either in a fixed manner (static) or dynamically. For example, a TV broadcaster uses the spectrum on a regular schedule, whereas a mobile operator may use spectrum dynamically based on demand.

Applications of IEEE 802.22 Standard

- 1. Rural and Remote Broadband Access
- **Internet Connectivity in Rural Areas**
- One of the most important applications of IEEE 802.22 is providing **broadband internet** to rural and remote areas where traditional broadband infrastructure (like DSL, cable, or fiber) is either unavailable or economically unfeasible to deploy. The long-range communication capabilities of IEEE 802.22, which can reach up to **100 kilometers**, make it ideal for extending internet access to hard-to-reach locations.

2. Smart City Applications

- **Internet of Things (IoT)**
- In **smart cities**, IEEE 802.22 can support the growing network of connected devices by providing an affordable and scalable communication infrastructure. Smart city applications such as **smart meters, traffic monitoring, environmental sensors, and smart lighting** require widespread, low-cost connectivity that can reach large areas. IEEE 802.22 is well-suited for supporting these applications with its long-range coverage and ability to connect a large number of devices.

Architecture of IEEE 802.22

- Base Station (BS)
- The **Base Station (BS)** is the central component in the network responsible for managing the communication of users (called **Customer Premises Equipment (CPE)**).
- The BS coordinates the transmission of data, handles spectrum management, and ensures the proper use of spectrum resources without interfering with primary users.

Customer Premises Equipment (CPE)

- **CPE** refers to devices at the users' locations that communicate with the base station. These devices can be laptops, smartphones, or any other type of broadband-enabled device. They are responsible for connecting to the base station and may also have cognitive radio capabilities to sense the available spectrum.

THE IEEE 802.22 Physical (PHY) Layer

- In IEEE 802.22, the Physical Layer (PHY) is responsible for ensuring robust and reliable communication over TV white spaces. The preamble, control header, and MAP (Medium Access Protocol) are key components for the operation of the PHY layer. These elements are used for synchronization, control, and managing the data transmission process in an efficient way.

THE IEEE 802.22 Physical (PHY) Layer

- **Preamble:**
- The preamble is used for synchronization and allows the receiver to properly align with the incoming signal. It helps in detecting the start of the frame and synchronizing time and frequency between the transmitter and the receiver.
- Functions of the Preamble:

Preamble

- **Timing Synchronization:** The preamble provides timing synchronization between the transmitter and receiver.
- **Frequency Synchronization:** It also aids in frequency synchronization, ensuring the receiver is tuned to the correct frequency.
- **Channel Estimation:** The preamble is used for estimating the channel conditions (e.g., multipath and fading effects), which helps in adjusting the modulation and coding schemes to match the channel quality.

Preamble

- **Frame Detection:** The preamble marks the beginning of a frame, signaling the start of a new transmission.
- **Protection Against Interference:** It helps the system avoid interference from other systems by providing enough time for the receiver to adjust.
- **Structure of the Preamble:**
 - The preamble is typically a short, fixed-length sequence that is easy to detect.
 - It usually contains a training sequence (used for channel estimation) and a delimitation pattern to mark the boundary between the preamble and the data payload.

Control Header

- The **control header** is responsible for carrying control information that helps in managing the communication between the transmitter and the receiver. It provides key metadata about the frame structure, transmission parameters, and other information needed to decode the rest of the frame.

Functions of the Control Header

- **Transmission Parameters:** It specifies key parameters for the transmission, such as modulation and coding schemes (MCS), frequency, and power levels.
- **Frame Type Identification:** The control header identifies the type of frame (e.g., data, acknowledgment, etc.)
- **MAC Addressing:** It may carry MAC addresses of the source and destination to route the data properly.
- **Error Detection:** The control header can include a checksum or CRC for error detection to ensure the integrity of the control information.

Structure of the Control Header

- **Header Fields:** The control header typically includes fields such as
- **Frame Control:** Defines the frame type, such as whether it is a data frame, acknowledgment, or a control frame.
- **Sequence Number:** Used to maintain the order of frames.
- **MAC Address:** The address of the destination and source devices.
- **Error Checking:** A cyclic redundancy check (CRC) or other error detection codes

MAP (Medium Access Protocol)

- The **MAP** is a critical component of the IEEE 802.22 standard, responsible for allocating resources for transmission. The MAP defines how resources are allocated within a time-frequency grid, specifying which user or device gets access to which resources at any given time.

Functions of the MAP(Medium Access Protocol)

- **Resource Allocation:** The MAP defines the allocation of time slots and subcarriers in the OFDM grid for each user or data flow. It indicates when and where data will be transmitted.
- **Frame Structure:** It provides the structure of the frame, defining the different parts of the frame, including data bursts, control information, and reserved slots.
- **Scheduling Information:** The MAP includes scheduling information that determines when each user or device should transmit or receive, based on factors like channel quality and fairness.

Structure of the MAP(Medium Access Protocol)

- The MAP structure contains several important components:
- **MAP Header:** Includes information such as the length of the MAP, the number of entries, and a checksum for error detection.
- **MAP Entries:** Each entry corresponds to a specific allocation, such as
- **Time Slot Allocation:** Specifies the time slots for transmission.
- **Frequency Subcarrier Allocation:** Defines which frequency subcarriers are assigned to the user or service.
- **Modulation and Coding Scheme (MCS):** Specifies the modulation and coding used for the allocated resources.
- **Transmission Power:** Specifies the transmission power for each user or device.

Preamble, Control Header, and MAP

Component	Purpose	Key Features
Preamble	Synchronization and detection of frame boundaries	Timing and frequency synchronization, channel estimation, frame detection, interference protection
Control Header	Carries metadata and control information for proper decoding	Includes transmission parameters, frame type, MAC addressing, error detection (e.g., CRC), and sequence numbers
MAP	Resource allocation and scheduling for transmission	Defines time and frequency allocation, scheduling information, burst details, modulation and coding schemes, and transmission power for users in the system

CBP(Control and Broadcast Packet)PACKET FORMAT

- The CBP (Control and Broadcast Packet) is part of the **Physical Layer** (PHY) structure used to carry control and broadcast messages within a WRAN(Wireless Regional Area Networks) cell. The main purpose of this packet is to facilitate communication between base stations and client devices (i.e., the WRAN devices).

CBP Packet Structure

- The **CBP packet format** at the PHY layer is typically structured as follows:
- **Preamble:** The preamble is used for synchronization, enabling the receiver to detect the packet and synchronize to the timing of the transmission. The length of the preamble is defined in the standard.

Header: The header of the CBP typically contains control information such as:

- **Packet Type** (e.g., control or broadcast).
- **Sequence Number** for tracking the packets.
- **Frame Control Information** for managing access to the medium.
- **Timing Information** (e.g., start and end times for synchronization).

CBP Packet Structure

- **Payload:** This part of the packet contains the actual data, which may include control information, network management data, or broadcast data for the WRAN network. The payload may be subdivided into smaller blocks for error correction and to meet the transmission requirements.
- **CRC:** The packet includes a Cyclic Redundancy Check (CRC) at the end for error detection. This ensures that any transmission errors can be identified.

CHANNEL CODING AND MODULATION SCHEMES IN IEEE 802.22

- **Channel Coding**
- Channel coding is used in IEEE 802.22 to protect data from errors during transmission over the wireless medium. The primary goal of channel coding is to add redundancy to the transmitted data so that the receiver can detect and correct errors that may occur due to noise, interference, or fading.

Turbo Codes

- The primary channel coding technique used in IEEE 802.22 is **Turbo coding**. Turbo codes are a class of high-performance error correction codes that provide strong protection against errors.
- Turbo codes are typically used for both the **downlink** (from the base station to the WRAN devices) and **uplink** (from WRAN devices to the base station) communications.
- These codes are based on **convolutional codes** and **interleaving**. The interleaving process helps to spread out the errors in time, allowing the decoder to better correct them.
- **Turbo encoding** involves two or more constituent codes (usually convolutional codes) and uses iterative decoding for higher reliability.

Reed-Solomon Codes

- In addition to turbo codes, **Reed-Solomon codes** are also used in IEEE 802.22, especially in the **physical layer** for Forward Error Correction (FEC).
- These codes are non-binary and are used to correct burst errors that occur due to interference and fading in the wireless channel.
- **Reed-Solomon** is particularly effective for correcting errors in bursts, which are common in wireless transmission, and it adds resilience to data integrity.

Modulation Schemes IN IEEE 802.22

- Modulation is the process of converting the data into a signal suitable for transmission over the wireless medium. In IEEE 802.22, a variety of modulation schemes are employed, depending on the channel conditions (such as signal-to-noise ratio, or SNR) and the data rate requirements.
- **QPSK (Quadrature Phase Shift Keying)**
- **16-QAM (Quadrature Amplitude Modulation)**
- **64-QAM**
- **BPSK (Binary Phase Shift Keying)**

QPSK (Quadrature Phase Shift Keying)

- QPSK is one of the most commonly used modulation schemes in IEEE 802.22, particularly in conditions with lower signal quality or when the channel SNR is low.
- In **QPSK**, two bits of data are encoded into one symbol, which allows the system to transmit at a relatively low bit rate but with robust performance.
- It's commonly used for **control channels** (e.g., CBPs) and **low-rate data transmissions**.

16-QAM (Quadrature Amplitude Modulation)

- **16-QAM** allows for a higher data rate than QPSK by encoding 4 bits of data into each symbol. This modulation scheme is used in scenarios where the channel conditions are better (higher SNR) and higher throughput is needed.
- It is generally used in **data channels** that can tolerate some error due to the higher complexity in error correction.
- **64-QAM:**
- **64-QAM** is used when even higher data rates are required and when the channel quality is sufficiently high to avoid too many errors in transmission.
- With **64-QAM**, each symbol represents 6 bits of data, which results in a higher data rate than both QPSK and 16-QAM. However, it is more sensitive to noise and interference.

BPSK (Binary Phase Shift Keying)

- **BPSK** is the most basic form of phase modulation, where one bit is encoded per symbol. This scheme is used in low SNR conditions, where the channel is very noisy, and robust communication is prioritized over high throughput.
- BPSK is typically used for **signaling** and **synchronization** purposes where error tolerance is minimal.

THE IEEE 802.22 MEDIUM ACCESS CONTROL(MAC) LAYER

- The **IEEE 802.22 Medium Access Control (MAC) layer** is a critical component of the **Wireless Regional Area Network (WRAN)** standard. The MAC layer in IEEE 802.22 is responsible for managing the medium access, ensuring fair and efficient use of the shared radio spectrum, and supporting communication between **base stations** and **client devices** (referred to as **subscriber stations** or SSs) in the WRAN.

MAC Superframe

- A MAC Superframe is typically used in low-power, wireless networks like Zigbee (IEEE 802.15.4), where devices communicate in a coordinated manner to save energy. The concept of a superframe helps organize time in a way that devices can schedule their activity and rest.
- **Superframe Structure:**
- **Beacon Period:** A portion of the superframe where a coordinator (or a network leader) sends out periodic beacons to synchronize the devices in the network.
- **Active Period:** During this period, devices in the network can transmit and receive data.
- **Inactive Period:** Devices may go to sleep or remain idle to conserve power.

MAC Frame

- A **MAC Frame** is a unit of data transmission in various protocols like IEEE 802.11 (Wi-Fi) and IEEE 802.15.4 (Zigbee). It serves to carry data and control information. Each MAC frame typically consists of the following fields
- **Frame Control**: Specifies the type of frame (data, acknowledgment, etc.), and other control information.
- **Sequence Number**: Used to track the frame's sequence to avoid confusion or duplicate frames.
- **Addressing Information**: This includes the source address, destination address, and other addressing elements.
- **Frame Payload**: The actual data or control information that's being transmitted.
- **FCS (Frame Check Sequence)**: A checksum used to detect errors in the transmitted frame.

Types of MAC Frames

- There are generally three main types of MAC frames:
- **Data Frames:** Carry data from one device to another.
- **Control Frames:** Used for managing and controlling access to the medium, including acknowledging received frames.
- **Management Frames:** Handle network management functions such as synchronization, association, and disassociation.

INCUMBENT DETECTION AND NOTIFICATION SUPPORT

- **Incumbent Detection and Notification Support in IEEE 802.22 MAC Layer** refers to mechanisms defined in the IEEE 802.22 standard, which is designed for **Wireless Regional Area Networks (WRANs)** operating in the **TV white spaces** spectrum (unused broadcast TV channels). The standard enables dynamic spectrum access to these "white spaces" while ensuring that the **incumbent users** (such as licensed TV broadcasters and other primary users) are protected from interference.

INCUMBENT DETECTION AND NOTIFICATION SUPPORT

- **Incumbent:** In the context of IEEE 802.22, the term "incumbent" refers to primary users (like TV broadcasters, meteorological services, etc.) that hold legal rights to operate within certain frequency bands.
- The IEEE 802.22 standard defines **mechanisms for detecting and avoiding interference to incumbent users**. These mechanisms involve both **incumbent detection** and **notification** features that allow secondary (unlicensed) users to coexist with primary users in shared frequency bands.

Incumbent Detection

- Incumbent detection ensures that secondary users (e.g., WRAN devices) are aware of the presence of incumbent users within a specific frequency band.
- The detection methods include:
- **Geolocation Database (or Spectrum Database):** Devices in the WRAN (secondary users) can use a geolocation database to determine the incumbent spectrum usage in their location. This helps identify the TV channels that are occupied by incumbents in the area.
- **Sensing (Radio Frequency (RF) Sensing):** WRAN devices can directly sense the presence of incumbent signals (e.g., broadcast TV signals) on the spectrum. If an incumbent signal is detected, the device will avoid transmitting on that channel to avoid causing interference.

Incumbent Detection

- **Signal Strength Thresholding:** The secondary devices can set thresholds for the detected incumbent signals. If the signal strength exceeds a certain threshold, it indicates the presence of an incumbent, and the secondary user must vacate the channel.
- **Cognitive Radio Technology:** IEEE 802.22 supports **cognitive radio** techniques, where WRAN devices can intelligently sense the spectrum and detect any incumbent transmissions in real time. The cognitive radio capabilities help the device make decisions about when and where to transmit.

Incumbent Notification

- When an incumbent user is detected (either through geolocation database queries or RF sensing), the IEEE 802.22 MAC layer provides notification mechanisms to inform the network and other devices. These notifications are important for maintaining interference-free operation and ensuring that the secondary users (WRAN devices) vacate the spectrum if necessary.
- **Notification to Other WRAN Devices:** Once an incumbent is detected, the device can notify other devices within the network so that they avoid using the same channels.
- **Communication with Database or Spectrum Manager:** When a secondary device detects an incumbent signal, it may notify the centralized **spectrum manager** or **database** that it has detected an incumbent in a particular frequency range. The spectrum manager can then adjust the available spectrum resources for other users in the area.
- **Dynamic Frequency Selection (DFS):** This is a part of the notification process where secondary devices dynamically select different frequency bands if the incumbent is detected. The system ensures that secondary users **move away from the detected incumbent** to a frequency band that is free from interference.

Multi-Channel Operation in IEEE 802.22

- **Multi-Channel Operation in IEEE 802.22** is a crucial feature that allows Wireless Regional Area Networks (WRANs) to efficiently utilize the available spectrum while avoiding interference with incumbent users, particularly in the TV white spaces. The **IEEE 802.22** standard is designed to enable unlicensed devices (secondary users) to access the unused portions of the TV broadcast spectrum, which are known as **TV white spaces**, in a dynamic and spectrum-efficient manner.
- **1. Dynamic Spectrum Access (DSA)**
- **2. Channel Availability and Incumbent Protection**
- **3. Frequency Planning and Coordination**
- **4. Multi-Channel Data Transmission**
- **5. Geolocation and Spectrum Database**
- **6. Channel Bonding**
- **7. Channel Access Mechanisms**
- **8. Quality of Service (QoS) and Load Balancing:**

Multi-Channel Operation in IEEE 802.22

- **Dynamic Spectrum Access (DSA):**
- IEEE 802.22 uses **Dynamic Spectrum Access (DSA)**, where secondary users (WRAN devices) can access multiple channels in the TV white space spectrum. This is essential for providing high bandwidth to users and maintaining efficient use of the spectrum. WRAN devices can operate across different channels within the available spectrum, which could include:
 - **TV Channels:** The channels available in the TV bands that are not currently being used by incumbent users (e.g., TV broadcasters).
 - **Channel Allocation:** The WRAN devices must dynamically choose the best channels based on real-time spectrum availability.

Multi-Channel Operation in IEEE 802.22

- **Channel Availability and Incumbent Protection:**
- To ensure that WRAN devices do not interfere with incumbent users, **incumbent detection** and **avoidance** mechanisms are built into the standard. These mechanisms allow WRAN devices to:
- **Detect incumbent users** (e.g., broadcast TV signals, wireless microphones, etc.) using methods like geolocation databases, RF sensing, and spectrum monitoring.
- **Vacate** the channels that are detected to be in use by incumbents.
- **Switch to available channels** that are free of interference from incumbent users, ensuring non-interference.

Multi-Channel Operation in IEEE 802.22

- **Frequency Planning and Coordination:**
- The IEEE 802.22 MAC layer is responsible for coordinating the use of multiple channels. The coordination mechanism ensures that multiple WRAN devices can operate simultaneously without causing interference to one another. Some of the key functions include:
 - **Channel Selection:** When a WRAN device (or the network) detects that a certain channel is occupied by an incumbent, it selects a different available channel from the spectrum pool.
 - **Channel Switching:** WRAN devices can switch between channels based on the availability of spectrum. The **MAC layer** plays a key role in managing this switching process to minimize service disruption.

Multi-Channel Operation in IEEE 802.22

- **Multi-Channel Data Transmission:**
- **Transmit and receive data on multiple channels** at once. This can significantly improve throughput and network efficiency.
- **Geolocation and Spectrum Database:**
- **Geolocation Databases:** WRAN devices use geolocation-based spectrum databases to determine which channels are available in a given area. This helps devices avoid interference with incumbent users by providing updated information on which TV channels are free

Multi-Channel Operation in IEEE 802.22

- **Channel Bonding:**
- **Bonded Channels:** These are combined channels that act as a larger single communication channel, thereby increasing the effective bandwidth available for data transmission.
- **Channel Access Mechanisms:**
- **Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA):** WRAN devices use this mechanism to avoid collisions when multiple devices attempt to access the same channel simultaneously. CSMA/CA is used to sense the channel for activity before transmission begins, ensuring no interference with other ongoing transmissions.

Multi-Channel Operation in IEEE 802.22

- **Quality of Service (QoS) and Load Balancing:**
- **QoS:** The multi-channel operation also supports the management of Quality of Service (QoS), ensuring that high-priority traffic (e.g., real-time applications like voice or video) gets appropriate access to the spectrum.
- **Load Balancing:** The system can dynamically balance the load across available channels to ensure that no single channel is overused while others are underutilized. This optimizes the network's overall performance.

SYNCHRONIZATION IN IEEE 802.22

- **Network Synchronization:**
- In a WRAN, synchronization refers to the alignment of time between different devices (e.g., secondary users, base stations, or WRAN devices). This ensures that devices know when to transmit and listen, preventing collisions and interference. It also enables power-saving mechanisms where devices can go into sleep mode during inactive periods.

SELF-COEEXISTENCE IN IEEE 802.22

- **Self-Coexistence** in the context of **IEEE 802.22** refers to the ability of **Wireless Regional Area Networks (WRANs)**, which operate in **TV white spaces**, to coexist without interference when multiple WRANs are operating in the same geographical area. Since **IEEE 802.22** is designed to allow unlicensed secondary users (WRAN devices) to access unused portions of the TV broadcast spectrum (white spaces), ensuring that these devices do not interfere with each other is a key aspect of the standard. This is particularly important because multiple WRANs might operate in nearby regions, and their operations need to be coordinated to avoid causing interference.

SPECTRUM MANAGEMENT MODEL IN IEEE 802.22

- **Dynamic Spectrum Access (DSA):** The IEEE 802.22 standard uses **dynamic spectrum access** to allow secondary users (WRAN devices) to opportunistically access vacant TV bands without causing interference to the primary users (licensed TV broadcasters). It continuously scans the spectrum and identifies available channels, ensuring that WRAN devices operate in unused portions of the spectrum.
- **Cognitive Radio Technology:** The WRAN system is built on **cognitive radio** technology, where devices have the ability to sense the spectrum, detect spectrum availability, and adapt their transmission parameters (such as frequency, power, and modulation) accordingly. This enables efficient spectrum utilization and avoids interference with primary users.

SPECTRUM MANAGEMENT MODEL IN IEEE 802.22

- **Spectrum Sensing and Coordination:** To ensure that secondary users do not interfere with primary users, the WRAN devices must periodically sense the spectrum to detect the presence of primary users (e.g., TV stations). **Spectrum sensing** is used to identify which frequencies are available.
- **Spectrum coordination** is then carried out by the base station (BS) to assign frequencies to different WRAN devices within a geographical region.
- **Geographical Database:** IEEE 802.22 relies on a **geographical database** maintained by a central authority. The database provides information on TV channels available for use in a particular region, taking into account the locations of primary users (TV stations), environmental conditions, and other factors that may affect spectrum availability.

SPECTRUM MANAGEMENT MODEL IN IEEE 802.22

- **Interference Management:** The standard also includes mechanisms for **interference management**, ensuring that when secondary users access the TV bands, their transmission does not interfere with the primary users. This involves power control, spectrum sharing rules, and coordination among WRAN devices to mitigate interference.

SPECTRUM SENSING

- spectrum sensing refers to the process of detecting and identifying the availability of unused or under-utilized portions of the radio spectrum, which is essential for dynamic spectrum access (DSA). This allows devices to utilize frequencies that are not being actively used by primary users (e.g., licensed spectrum holders), without causing harmful interference.
- **Spectrum Holes:** These are frequency bands that are temporarily unused or underutilized by the primary (licensed) users. Spectrum sensing helps identify these gaps so secondary (unlicensed) users (such as sensors or cognitive radios) can transmit data without interference.

SPECTRUM SENSING

- **Types of Spectrum Sensing:**
- **Energy Detection:** Measures the energy present in a particular frequency band. If the energy exceeds a threshold, it is assumed that the band is occupied. This method is simple and does not require prior knowledge of the signal, but it can be affected by noise and interference.
- **Matched Filtering:** Requires knowledge of the primary user's signal and compares it with a known reference signal. It offers high detection accuracy but requires detailed information about the primary signal, which may not always be available.

SPECTRUM SENSING

- **Cyclostationary Feature Detection:** This method exploits the periodicity or cyclostationary characteristics of signals (such as modulation patterns) to detect the presence of primary users. It provides a good balance between detection accuracy and the ability to operate in noisy environments.
- **Waveform-Based Detection:** This method involves recognizing specific waveforms or patterns used by primary users to detect spectrum occupancy. It is more complex but also more robust.

TCP protocol overview

- **Transmission Control Protocol (TCP)** is one of the core protocols of the **Internet Protocol Suite** (often referred to as TCP/IP), which is the foundation for internet communication. TCP is a **connection-oriented** and **reliable** protocol that ensures data is delivered between computers in a consistent and accurate manner.

DESIGNED AND FINE-TUNED TO WIRED NETWORKS

- In **wired networks**, the transmission quality is generally reliable, and latency is predictable. However, **wireless sensor networks** often experience **high latency** due to intermittent connectivity, multi-hop communication, and unreliable links, which can lead to **time-outs** and retransmissions in TCP.

TCP BASICS

- **Transmission Reliability:**
- Reliability is maintained but with **local retransmissions** and **link-layer feedback** instead of end-to-end retransmissions.
- **Flow Control:**
- Flow control can be implemented with **adaptive sliding window techniques** based on the available resources at the receiver and the current network conditions.
- **Congestion Avoidance:**
- Congestion avoidance mechanisms are often **relaxed** in sensor networks, as the main issue is often link reliability, not network congestion. Some adaptations focus on **minimizing retransmission attempts** for packet loss.

TCP HEADER FORMAT

- The **standard TCP header** consists of 20 bytes of fixed-size fields and optional fields. Here is the general structure of the **TCP header**:

Field	Size (bits)	Description
Source Port	16	Port number of the sender
Destination Port	16	Port number of the receiver
Sequence Number	32	Identifies the sequence of the sent data byte
Acknowledgment Number	32	If ACK flag is set, this is the next expected byte from the receiver
Data Offset (Header	4	Specifies the length of the TCP header (in 32-bit words)
Reserved	3	Reserved for future use,

CONGESTION CONTROL

- **Congestion control** is a critical aspect of the **Transmission Control Protocol (TCP)** that ensures efficient data transfer across networks by preventing congestion. In the context of TCP, **congestion** refers to the situation where the network's resources (such as bandwidth, buffer space in routers, etc.) are overwhelmed by excessive traffic, which can result in packet loss, delays, and reduced overall network performance. TCP uses a set of algorithms to **detect**, **avoid**, and **mitigate** congestion, ensuring smooth and reliable communication between sender and receiver.

CONGESTION CONTROL

- TCP uses four key algorithms to manage congestion:
- **1. Slow Start**
- **2. Congestion Avoidance**
- **3. Fast Retransmit and Fast Recovery**
- **4. Timeout and Recovery**

1.Slow Start

- **Slow Start** is a mechanism that controls the initial phase of a TCP connection when it begins sending data. At the start of the connection (or after a timeout), TCP begins with a small **congestion window (cwnd)**, typically set to one or two Maximum Segment Size (MSS) units.

2. Congestion Avoidance

- Once the **congestion window (cwnd)** exceeds a certain threshold (the **ssthresh**), TCP switches from **slow start** to **congestion avoidance**. In this phase, TCP stops growing the **cwnd** exponentially and instead increases it more gradually.

3. Fast Retransmit and Fast Recovery

- When packet loss occurs due to congestion (usually indicated by the lack of an acknowledgment), TCP triggers **fast retransmit** and **fast recovery** to quickly recover from packet loss without resorting to a full **timeout**.

4. Timeout and Recovery

- When a packet loss is severe and TCP doesn't receive any acknowledgment for the sent data (meaning no duplicate ACKs are received), the sender assumes that the packet was lost due to **congestion** and enters **timeout recovery**.

ROUND-TRIP TIME ESTIMATION

- In the **TCP (Transmission Control Protocol)**, **Round-Trip Time (RTT)** refers to the time it takes for a data packet to travel from the sender to the receiver and back again (i.e., from the sender to the receiver and the acknowledgment (ACK) being returned to the sender). **RTT estimation** is critical in TCP because it helps to determine the retransmission timeout (RTO) and plays an important role in various aspects of TCP's performance, such as congestion control, flow control, and overall responsiveness.

ROUND-TRIP TIME ESTIMATION

- RTT estimation works step-by-step:
- **1. Initial RTT Measurement:** The sender measures the RTT when the first few packets are sent and receives the corresponding ACKs.
- **2. Smoothing RTT:** The sender calculates the **SRTT** and **RTTVAR** based on the observed RTTs, using the smoothing formulas.
- **3. RTO Calculation:** The sender uses the **SRTT** and **RTTVAR** to compute the **RTO**. The **RTO** determines the timeout interval for waiting for an ACK.
- **4. Retransmission:** If the sender does not receive an ACK before the **RTO** expires, it retransmits the data, triggering the **fast retransmit** mechanism and adjusting the congestion window accordingly.
- **5. Continuous Update:** Throughout the communication, TCP continues to adjust the **SRTT**, **RTTVAR**, and **RTO** dynamically based on measured RTTs and the variation in those values.

SOLUTIONS FOR TCP OVER ADHOC

1.MOBILITY RELATED

- TCP feedback in the context of mobility, particularly in Mobile Ad Hoc Networks (MANETs) or wireless environments, is essential for adapting the congestion control and error recovery mechanisms in TCP to deal with the challenges that arise from node mobility. Mobility introduces frequent route changes, varying link quality, disconnections, and handovers, all of which can cause packet losses and delays, complicating the way TCP handles congestion and feedback.
- 1.TCP –FEEDBACK
- 2.THE ELFN(EXPLICIT LINK FAILURE NOTIFICATION) APPROACH
- 3.FIXED RTO
- 4.THE ATCP PROTOCOL
- 5.TCP - DOOR

TCP –FEEDBACK

- TCP relies heavily on feedback from the receiver to regulate the flow of data from the sender.
- The two main types of feedback used by TCP are:
- **Acknowledgments (ACKs)**: The receiver sends an acknowledgment (ACK) for successfully received packets back to the sender. This helps the sender know which packets have been successfully delivered and which have been lost.
- **Congestion Window (CWND)**: The sender adjusts its congestion window based on feedback received from ACKs. When ACKs are received in a timely manner, the sender increases the window size to send more data. When packet loss or congestion is inferred from the lack of ACKs (timeouts or duplicate ACKs), the sender reduces its transmission rate.

THE ELFN(EXPLICIT LINK FAILURE NOTIFICATION) APPROACH

- Understanding TCP's behavior in ad hoc networks is crucial due to the dynamic nature of these environments. Traditional TCP, designed for wired networks, often performs poorly in ad hoc settings because it interprets link failures as congestion, leading to unnecessary and detrimental congestion control mechanisms. This is where the ELFN (Explicit Link Failure Notification) approach comes
- Problem:
- Ad hoc Network Characteristics:
 - 1.Frequent topology changes due to node mobility.
 - 2. Link failures are common and often not due to congestion.
 - 3. Wireless interference and varying channel conditions.
- The ELFN Solution:
- ELFN introduces a mechanism for nodes to explicitly notify the TCP sender when a link failure occurs

FIXED RTO(retransmission timeout)

- Problem with Standard TCP RTO:
- Standard TCP uses a retransmission timeout (RTO) to determine when to retransmit a lost packet.
- When a timeout occurs, TCP increases the RTO, often exponentially, to avoid overwhelming a congested network.
- Fixed RTO Solution:
- The "fixed RTO" approach aims to distinguish between packet loss caused by congestion and packet loss caused by link failures.
- When consecutive retransmission timeouts occur, the TCP sender assumes that a link failure has happened.
- The RTO remains fixed until a new route is established and the lost packet is acknowledged.

THE ATCP(Ad hoc TCP) PROTOCOL

- The ATCP (Ad hoc TCP) protocol is a solution designed to improve the performance of TCP in mobile ad hoc networks (MANETs).
- Layered Approach:
- ATCP implements a thin layer between the TCP and IP layers in the protocol stack.
- This allows ATCP to monitor network conditions and provide feedback to TCP without requiring modifications to the core TCP protocol itself.

TCP - DOOR

- TCP-DOOR" refers to a technique that focuses on detecting and responding to out-of-order (OOO) packet delivery. This is particularly relevant in mobile ad hoc networks (MANETs) where dynamic routing can lead to packets arriving at the destination in a sequence different from how they were sent.
- **Out-of-Order Delivery:**
- In MANETs, routes can change frequently due to node mobility.
- This can cause packets to take different paths and arrive at the destination out of order.
- **TCP-DOOR's Solution:**
- Detection of Out-of-Order Delivery:
- TCP-DOOR aims to accurately detect when out-of-order delivery occurs.

TCP - DOOR

- Timestamping:
- Often variations of TCP-DOOR will include time stamping of packets, which allow for a more accurate assessment of the order of packets, and therefore a more accurate detection of out of order packets.

SOLUTIONS FOR TCP OVER ADHOC

2.FAIRNESS -RELATED

- **fairness in TCP (Transmission Control Protocol) over ad hoc networks**, we are dealing with challenges arising from the unique characteristics of these networks, such as:
- **Dynamic topology**: Nodes frequently move and connections constantly change.
- **Limited bandwidth and resources**: Bandwidth can be shared inefficiently, and resources like battery and power are scarce.
- **Multihop communication**: Packets often have to travel multiple hops to reach the destination, which introduces delays, jitter, and potential for loss.

SOLUTIONS FOR TCP OVER ADHOC

2.FAIRNESS -RELATED

- 1.COPAS
- 2.NEIGHBORHOOD RED

COPAS

- fairness in TCP over ad hoc networks, particularly concerning the "capture effect," the COPAS (COntention-based PAth Selection) algorithm is a significant contribution.
- **The "Capture Effect" and Fairness:**
- In wireless ad hoc networks, the "capture effect" occurs when nodes with a stronger signal or those closer to a bottleneck link dominate the shared wireless medium. This leads to unfair bandwidth allocation, where some TCP flows are severely starved while others thrive.
- This unfairness significantly degrades overall network performance and user experience.

COPAS

- COPAS and its Fairness Contribution:
- **Disjoint Path Routing:**
- COPAS selects multiple, node-disjoint routes between the source and destination. This helps distribute traffic and reduce contention at bottleneck links.
- **Dynamic Contention Balancing:**
- COPAS monitors and dynamically adjusts routing paths based on the level of MAC layer contention. This allows the network to adapt to changing conditions and avoid persistent capture situations.

NEIGHBORHOOD RED

- "Neighborhood RED" (Neighborhood Random Early Detection) in the context of fairness in TCP over ad hoc networks addresses the limitations of traditional RED (Random Early Detection) in these dynamic environments.
- **The Problem with Traditional RED in Ad Hoc Networks:**
- Fairness Concerns:
- In ad hoc networks, nodes with more neighbors or those closer to bottleneck links may experience higher queue lengths and therefore be more likely to have packets dropped by RED.
- This can lead to unfair bandwidth allocation, where some nodes are penalized more than others.

NEIGHBORHOOD RED

- Neighborhood RED's Solution:
- Neighborhood Awareness:
- Neighborhood RED extends traditional RED by incorporating information about the node's neighborhood.
- This allows the algorithm to consider the congestion levels of neighboring nodes when making packet dropping decisions.